

Petrochemical Technology

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[Review of Latest Petrochemical Processes]

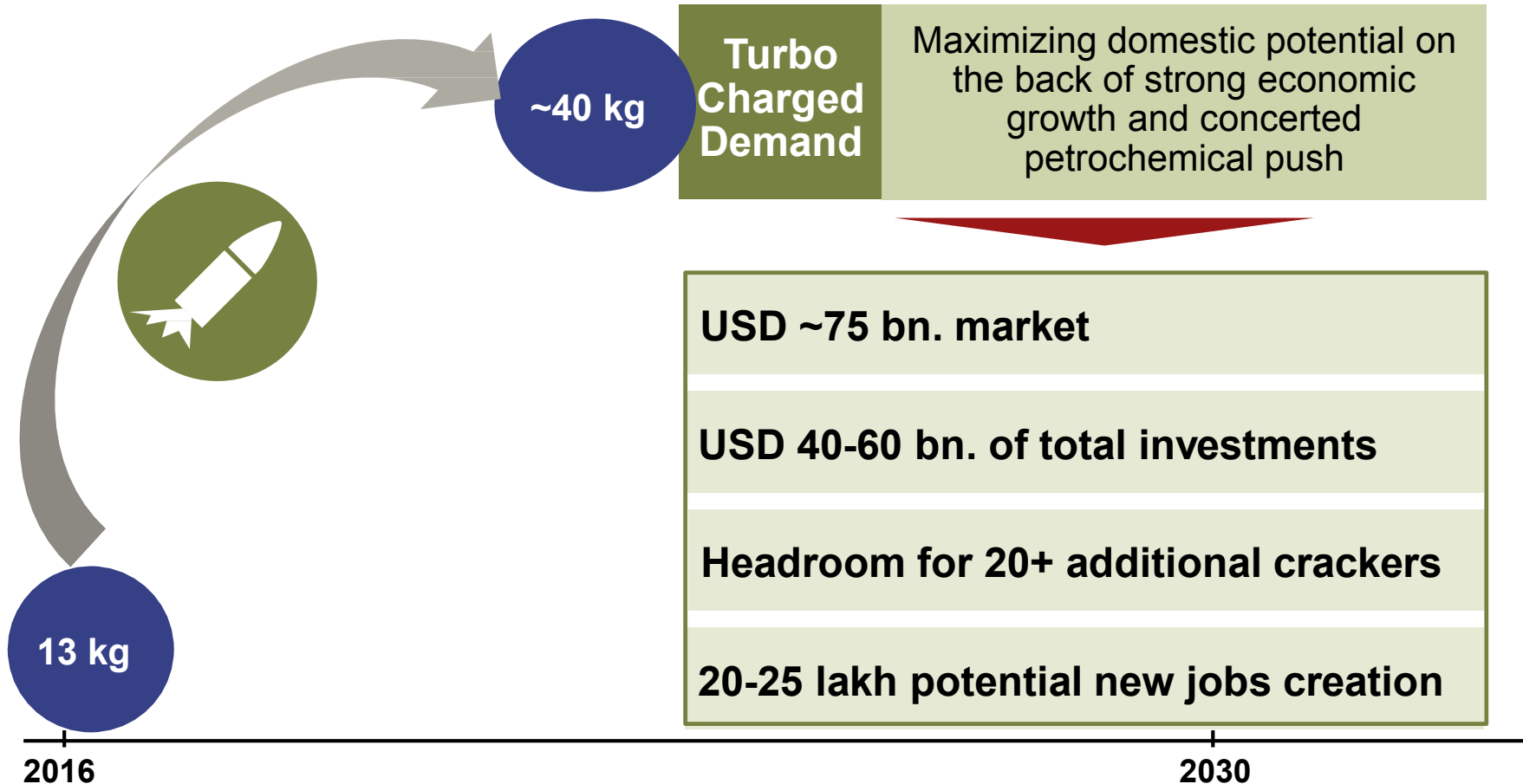
Petrochemical Sector – a key pillar of the economy

Petrochemicals are commercially used chemicals derived from various chemical compounds, mainly hydrocarbons, which are again derived from crude oil (petroleum) and natural gas.

Petrochemicals play a vital role in economic development & growth of a country. The growth of this industry is closely linked to economic growth of a country. Petrochemicals are considered as enablers for growth of other sectors of the economy. Today, petrochemical products permeate the entire spectrum of daily use items and cover almost every sphere of life like clothing, housing, construction, furniture, automobiles, household items, agriculture, horticulture, irrigation, packaging, medical appliances, electronics and electrical, etc.

Indian petrochemical demand can register 3X growth - potential to be a ~USD \$75 billion market by 2030

India's per capita consumption growth



Per Capita Primary Energy Consumption International Comparisons: 2021

(Figures in Kg. Oil Equivalent)

S. No.	Country	Oil	Natural Gas	Coal	Nuclear Energy	Hydro electricity	Renewables	Total	Population (Million) Mid-2021
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
1	China	503	230	1456	62	207	191	2650	1412
2	US	2467	2138	759	532	174	537	6607	332
3	India	157	38	344	7	26	31	603	1393
4	Russian Federation	1067	2797	559	329	331	10	5093	146
5	Japan	1223	710	913	105	139	250	3341	125
6	Canada	2533	2680	300	519	2241	361	8634	38
7	Germany	1168	936	608	179	52	655	3597	83
8	South Korea	1471	632	853	401	8	123	3489	85
9	Brazil	486	163	80	15	383	268	1395	213
10	Iran	1454	3999	34	15	65	8	5575	52
11	Saudi Arabia	4259	2807	2	-	-	5	7074	36
12	France	1033	566	85	1253	200	270	3408	65
13	Indonesia	239	116	285	-	20	55	714	275
14	United Kingdom	858	979	74	147	17	439	2514	68
15	Turkey	340	382	323	-	97	113	1255	129
16	Mexico	709	905	66	31	93	110	1913	84
17	Italy	923	1056	93	-	164	308	2544	59
18	Australia	1736	1312	1508	-	139	545	5242	26
19	Spain	852	437	58	183	100	347	1976	67
20	Thailand	1099	852	407	-	22	161	2541	47
21	South Africa	402	55	1402	37	5	63	1965	60
22	Taiwan	1893	998	1701	255	33	122	5001	24
23	United Arab Emirates	4290	6085	180	232	-	118	10905	10
24	Poland	975	609	1370	-	16	235	3204	33
25	Vietnam	217	61	511	-	170	64	1023	100
26	Malaysia	884	924	555	-	190	39	2593	38
27	Pakistan	231	376	157	33	83	13	893	102
28	Egypt	132	236	5	-	15	10	398	225
29	Netherlands	6159	5286	982	145	4	1797	14372	6
30	Singapore	3886	656	44	-	-	17	4603	18
Rest of World		285	241	86	30	94	42	779	2487
Total World		545	443	488	77	123	122	1797	7837

World Energy Outlook-2021

Total Primary Energy Consumption (Exajoules)		Refining Capacity (Thousand barrels daily)	
World	595	World	101912
India (Rank 3)	35	India (Rank 4)	5018
Top Five Countries		Top Five Countries	
China	158	US	17941
United States of America	93	China	16990
India	35	Russian Federation	6861
Russian Federation	31	India	5018
Japan	18	South Korea	3572
Production - Oil (Thousand barrels daily)		Consumption- Oil (Thousand barrels daily)	
World	89877	World	94088
India (Rank 21)	746	India (Rank 3)	4878
Top Five Countries		Top Five Countries	
US	16585	US	18684
Saudi Arabia	10954	China	15442
Russian Federation	10944	India	4878
Canada	5429	Saudi Arabia	3595
Iraq	4102	Russian Federation	3407
Production - Natural Gas (BCM)		Consumption- Natural Gas (BCM)	
World	4037	World	4037
India (Rank 26)	29	India (Rank 14)	62
Top Five Countries		Top Five Countries	
US	934	US	827
Russian Federation	702	Russian Federation	475
Iran	257	China	379
China	209	Iran	241
Qatar	177	Canada	119

Crude Oil Imports (Thousand barrels daily)		Crude Oil Exports (Thousand barrels daily)	
World	42738	World	41228
India (Rank 3)	4244	India	-
Top Five Countries		Top Five Countries	
China	10301	Saudi Arabia	6227
United States	6110	Russia	4510
India	4244	Iraq	3440
South Korea	2613	Canada	3195
Japan	2493	United States	2980
Petroleum Product Imports (Thousand barrels daily)		Petroleum Product Exports (Thousand barrels daily)	
World	27796	World	29370
India (Rank 6)	1013	India (Rank 7)	1283
Top Five Countries		Top Five Countries	
Singapore	1977	United States	5652
United States	2359	Russia	2385
Netherlands	1745	Netherlands	2258
China	1428	Singapore	1426
South Korea	1080	Saudi Arabia	1344
Natural Gas Imports (BCM)		Natural Gas Exports (BCM)	
World	1411	World	1419
India (Rank 15)	31	India	-
Top Five Countries		Top Five Countries	
Germany	154	Russia	247
China	143	United States	188
Japan	105	Qatar	135
United States	80	Norway	111
Italy	73	Australia	107

Petrochemicals

**Total Petrochemicals Production : 44.588 MMT]
During 2021-22**

**A: Intermediates
[22.686 MMT]**

Building Blocks

Olefins (3): Ethylene, Propylene and Butadiene
[12.527 MMT]

Aromatics (5): Benzene, Toluene, *ortho*-Xylene,
[4.677 MMT] *para*-Xylene and Mixed Xylene

Fibre Intermediates (4) [5.482 MMT] :

Acrylonitrile, Caprolactum, Mono Ethylene Glycol (MEG),
and Purified Terephthalic Acid (PTA)

B: Major Petrochemicals [19.372 MMT]

Synthetic Fibres (10) [4.04 MMT] :

Acrylic Fibre, Polyester Staple Fibrefill, Nylon Filament Yarn, Nylon Industrial Yarn/Tyre Cord, Polyester Filament Yarn, Polyester Staple Fibre, Polypropylene Filament Yarn, Polypropylene Staple Fibre, Polyester Industrial Yarn, Elastomeric/Spandex Filament Yarn

Polymers (7) [12.47 MMT] :

Linear Low Density Polyethylene, Low Density Polyethylene, High Density Polyethylene, Polystyrene, Polypropylene (Inc. Co- Polymer), Expandable Polystyrene, Poly Vinyl Chloride

Synthetic Rubber (Elastomers) (4) [0.382 MMT] :

Styrene Butadiene Rubber (SBR), Poly Butadiene Rubber (PBR), Nitrile Butadiene Rubber (NBR), Ethyl Vinyl Acetate (EVA)

Performance Plastics (8) [1.70 MMT] :

Acrylonitrile Butadiene Styrene (ABS) Resin, Nylon-6, Nylon 6,6, Polymethyl Methacrylate (PMMA), Styrene Acrylonitrile (SAN) Resin, Polytetrafluoroethylene (PTFE), Polyester Chips/PET Chips, Polycarbonate

Synthetic Detergent Intermediates (2) [0.78 MMT] :

Linear Alkyl Benzene (LAB), Ethylene Oxide (EO)

C: Other Petro-based Chemicals (19)

[2.53 MMT]

Diethylene Glycol, Diacetone Alcohol, Ethylene Dichloride, Butanol, 2-Ethyl Hexanol, **Vinyl Chloride Monomer**, PBT, Polycarbonate, **Propylene Oxide**, **Propylene Glycol**, Polyvinyl Acetate Resin, Unsaturated Polyester Resin, Methyl Methacrylate, Iso-Butanol, C4-Raffinate, **Phthalic Anhydride**, Vinyl Acetate Monomer, Isopropanol, Polyol.

Olefinic Complexes

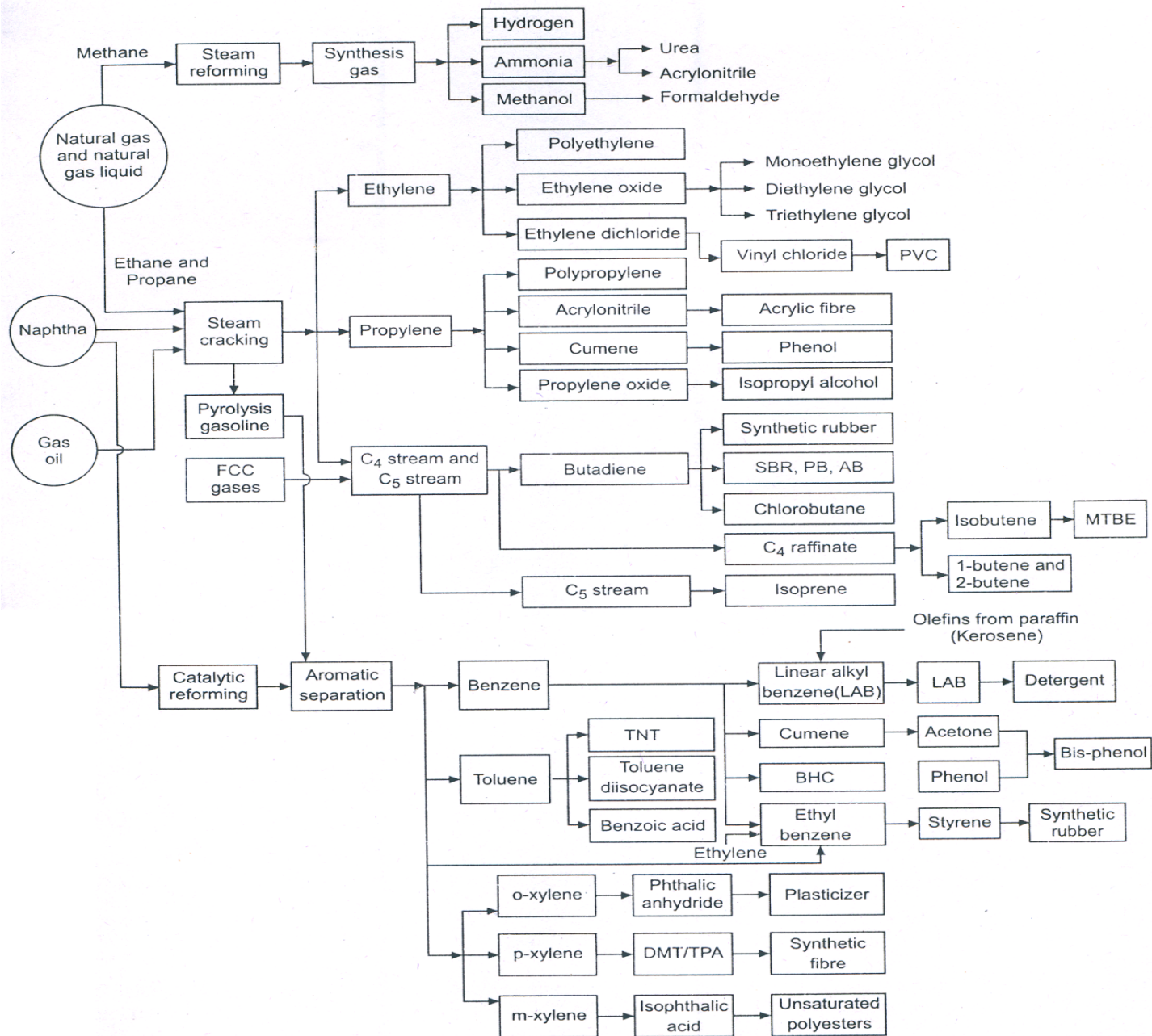
Presently there are eleven crackers, in operation with combined ethylene capacity of about 7.05 million tonnes per annum as given below:

Naphtha Crackers	Ethylene Capacity (MTs)
Reliance Industries Ltd., Vadodara, Gujarat	1,75,000
Reliance Industries Ltd, Hazira, Gujarat	8,87,000
Haldia Petrochemicals Ltd, Haldia, West Bengal	7,00,000
Indian Oil Corporation Ltd., Panipat, Haryana	8,00,000
Sub- Total	25,62,000
Natural Gas Crackers	
Reliance Industries Ltd., Nagothane, Maharashtra	4,21,400
Gas Authority of India Limited, Pata, UP	8,50,000
Reliance Industries Ltd, Dahej, Gujarat	4,00,000
Sub- Total	16,71,400
Dual Feed Crackers	
Brahmaputra Cracker and Polymer Limited (BCPL)	2,13,100
ONGC Petro additions Ltd. (OPaL)*	11,00,000
Sub- Total	13,13,100
Off Gas Cracker	
Reliance Gas industry, Jamnagar	15,00,000
Sub- Total	15,00,000
Total	70,46,500

There are six aromatic complexes in the country with a combined xylene capacity of 5.5 MMT

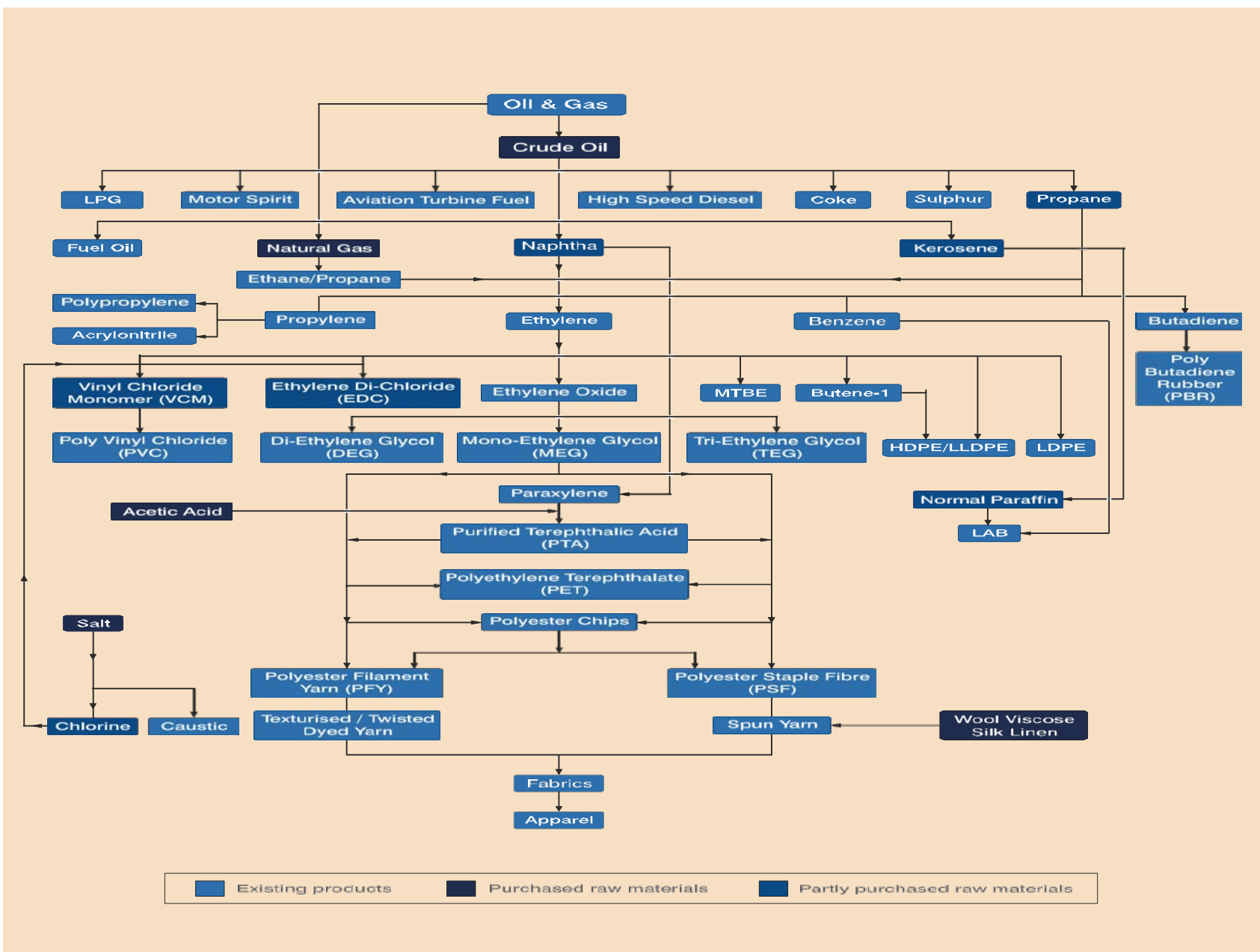
There are six Petro Fluid Catalytic Cracking Units with a combined propylene capacity of 1.7 MMT

Paradip Refinery: Ethylene potential – 200 KTA
Propylene potential – 700 KTA



Overall View of a Petrochemical Complex

Product Flow in RIL's Integrated Petrochemical Complex



Domestic and Overseas Production of Oil & Gas

Entity/Year	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22 (P)
(1)	(2)	(3)	(4)	(5)	(6)	(7)
Domestic Production of Oil & Gas						
Total Production of Oil & Condensate (MMT)	36.01	35.68	34.20	32.17	30.49	29.69
Total Production of Gas (BCM)	31.90	32.65	32.87	31.18	28.67	34.02
Total Domestic O & OEG Production (MMTOE)	67.91	68.33	67.08	63.35	59.17	63.71
Overseas Production of Oil & Gas						
3. Overseas Production of Crude Oil & Condensate (MMT)						
1. ONGC Videsh Ltd. (OVL)	8.43	9.35	10.10	9.76	8.51	8.10
2. Bharat PetroResources Ltd. (BPRL)	0.79	1.53	2.15	2.07	1.79	1.87
3. Indian Oil Corporation Ltd. (IOCL)	1.15	2.68	3.24	3.21	2.71	2.88
4. GAIL (India) Ltd. (GAIL)	0.05	0.06	0.06	0.04	0.02	0.02
5. OIL (Oil India Ltd.)	0.91	1.63	1.69	1.58	1.42	1.40
6. HPCL-PPCL (Prize Petroleum Co. Ltd.)	0.01	0.01	0.01	0.00	0.00	0.003
Total-Oil Production	11.34	15.26	17.25	16.66	14.45	14.27
4. Overseas Production of Natural Gas (BCM)						
1. ONGC Videsh Ltd. (OVL)	4.37	4.81	4.74	5.23	4.53	4.23
2. Bharat PetroResources Ltd. (BPRL)	0.35	0.65	0.68	0.60	0.67	0.80
3. Indian Oil Corporation Ltd. (IOCL)	0.78	1.00	1.05	0.95	1.07	1.28
4. GAIL (India) Ltd. (GAIL)	0.45	0.44	0.37	0.55	0.50	0.46
5. OIL (Oil India Ltd.)	0.35	0.71	0.70	0.62	0.68	0.81
6. HPCL-PPCL (Prize Petroleum Co. Ltd.)	0.05	0.05	0.05	0.03	0.03	0.02
Total Gas Production	6.35	7.66	7.58	7.98	7.48	7.60
Total Overseas Oil & OEG Production						
1. ONGC Videsh Ltd. (OVL)	12.80	14.16	14.83	14.98	13.04	12.33
2. Bharat PetroResources Ltd. (BPRL)	1.14	2.18	2.83	2.68	2.46	2.66
3. Indian Oil Corporation Ltd. (IOCL)	1.93	3.68	4.29	4.15	3.78	4.17
4. GAIL (India) Ltd. (GAIL)	0.51	0.49	0.43	0.59	0.47	0.44
5. OIL (Oil India Ltd.)	1.26	2.34	2.39	2.20	2.10	2.20
6. HPCL-PPCL (Prize Petroleum Co. Ltd.)	0.06	0.06	0.06	0.04	0.04	0.02
Total Overseas Oil & OEG Production (in MMTOE)	17.70	22.92	24.83	24.64	21.88	21.83
Overseas production as (%) percentage of domestic production	26.06	33.55	37.02	38.89	36.99	34.26

Basin-wise Production of Crude Oil

(Figures in TMT)

Basin	State	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22 (P)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	
Crude Oil Production by Oil and Natural Gas Corporation							
Western Offshore*	Maharashtra	16268	16231	14950	14517	14236	13604
Krishna Godavari Offshore (Eastern offshore)	Andhra Pradesh	16	8	18	19	13	14
Western Onland	Gujarat	4468	4452	4492	4574	4484	4388
Assam & Assam Arakan	Assam	953	975	993	987	964	994
Krishna Godavari	Andhra Pradesh	276	322	285	202	154	159
Cauvery	Tamilnadu	238	262	303	328	331	291
ONGC-Nomination		22218	22249	21042	20627	20183	19451
ONGC-NELP/JVs Share		3316	3186	3189	2726	2350	2256
Total ONGC's Production		25534	25435	24231	23353	22533	21707
Crude Oil Production by Oil India Ltd.							
Assam-Arakan	Assam	3250	3367	3282	3075	2905	2957
Assam-Arakan	Arunachal Pradesh	8	7	9	23	24	21
Rajasthan	Rajasthan	0	2	3	8	9	9
OIL-Nomination		3258	3376	3293	3107	2937	2987
OIL-JVs Share		19	17	14	26	27	23
Total OIL's Production		3277	3393	3307	3133	2964	3010
Crude Oil Production by Pvt/JVs-in PSC + RSC/CBM							
Krishna Godavari Offshore (Eastern offshore)	Andhra Pradesh	960.2	814.5	633.7	537.1	731	611.8
Cauvery Offshore (Eastern offshore)	Tamilnadu	2.4	4.2	2.2	0.7	0	0.0
Cambay (Offshore)	Gujarat Offshore	381.6	399.1	735.6	590.8	411	338.2
Mumbai Offshore (Western Offshore)	Western Offshore	792.5	687.2	527.4	337.8	0	0.0
KG Onshore	Andhra Pradesh	0.0	0.0	11.2	41.2	40	43.0
Assam-Arakan	Arunachal Pradesh	47.6	42.9	34.1	32.2	30	26.8
	Assam	0.0	3.1	34.4	30.1	33	40.0
Cambay	Gujarat	137.2	139.3	133.5	133.8	167	238.6
Rajasthan	Rajasthan	8164.6	7885.3	7663.8	6644.9	5882	5878.7
Cauvery	Tamilnadu	46.3	83.1	92.1	87.5	79	75.5
Bengal	West Bengal	0.0	0.0	0.0	0.0	0	0.1
Total Pvt/JVs Production-in PSC + RSC/CBM **		10532	10059	9868	8436	7374	7253
Total Crude Oil Production		36009	35684	34203	32170	30494	29691

Basin-wise Production of Natural Gas

(Figures in MMSCM)

State	Basin	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22 (P)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	
Natural Gas Production by Oil and Natural Gas Corporation							
Western Offshore	Maharashtra	16413	17243	18005	17645	16660	15,681
Krishna Godavari Offshore (Eastern offshore)	Andhra Pradesh	470	548	1037	931	426	262
Western Onland	Gujarat	1490	1532	1332	1274	1074	951
Frontier	Rajasthan	5	6	4	0	0	
Assam & Assam Arakan	Assam	435	508	483	460	387	373
	Tripura	1430	1440	1554	1473	1634	1531
Krishna Godavari	Andhra Pradesh	868	959	1077	897	814	794
Cauvery	Tamilnadu	977	1194	1182	1066	877	1037
ONGC-Nomination		22088	23429	24675	23746	21872	20629
ONGC-NELP/JVs Share		1182	1180	1136	1150	944	1,051
Total ONGC's Production		23270	24610	25810	24896	22816	21680
Natural Gas Production by Oil India Ltd.							
Assam-Arakan	Assam	2693	2659	2482	2379	2241	2620
Assam-Arakan	Arunachal Pradesh	12	12	12	38	50	53
Rajasthan	Rajasthan	232	211	228	251	188	219
OIL-Nomination		2937	2881	2722	2668	2480	2893
OIL-JVs Share		0	23	143	133	162	153
Total OIL's Production		2937	2905	2865	2801	2642	3045
Natural Gas Production by Pvt/JVs-in PSC + RSC/CBM							
Krishna Godavari Offshore (Eastern offshore)	Andhra Pradesh	3183	2245	1338	849	1182	6761
Cauvery Offshore (Eastern offshore)	Tamilnadu	18	24	84	43	13	2
Cambay (Offshore)	Gujarat Offshore	81	98	122	129	148	163
Mumbai Offshore (Western Offshore)	Maharashtra	1874	1853	1531	1039	0	0
KG Onshore	Andhra Pradesh			5	15	13	15
Assam-Arakan	Arunachal Pradesh	16	18	16	7	5	5
	Assam		53	324	302	367	378
Cambay	Gujarat	90	75	70	69	65	66
Rajasthan	Rajasthan	1041	1225	1251	1632	1852	2399
Cauvery	Tamilnadu	6	14	25	31	34	29
Ranigunj South (CBM)	West Bengal	170	203	153	139	110	112
Ranigunj East (CBM)	West Bengal	385	328	198	167	197	277
Jharia (CBM)	Jharkhand	3	4	4	5	2	4
Sohagpur East / West (CBM)	Madhya Pradesh	6	200	357	345	334	290
Total Pvt/JVs Production-in PSC + RSC/CBM		6872	6338	5477	4770	4321	10502
Total Natural Gas Production		31897	32649	32873	31184	28673	34024

Status of LNG Terminals in India

Terminal	Promoters	Capacity (MMT- PA) as on 31.03.2022	Capacity Utilization (%)					
			2016- 17	2017- 18	2018- 19	2019- 20	2020- 21	2021- 22
(1)	(2)	(3)	(4)	(5)	(6)	(7)		(8)
Operational LNG Terminal								
Dahej (Gujarat)	Petronet LNG Ltd. (PLL)	17.50	105.31	105.30	106.50	102.08	93.69	87.50
Kochi (Kerala)		5.00	5.46	12.20	9.30	16.59	18.03	20.72
Hazira (Gujarat)	Hazira LNG Pvt. Ltd. (HLPL)	5.00	69.63	58.90	79.90	94.59	76.80	47.20
Dabhol (Maharashtra) ¹	Ratnagiri Gas and Power Pvt. Ltd. (RGPPL- JV of GAIL & NTPC)	2.9*	57.07	64.90	72.10	97.40	75.44	85.15
Ennore (Tamilnadu)	IndianOil LNG Pvt. Ltd. (IOLPL)	5.00	-	-	-	9.00	13.00	14.00
Mundra (Gujarat)	GSPC LNG Limited (GLL)	5.00	-	-	-	29.63	34.80	19.30
Planned/Under-Construction LNG Terminals								
Dhamra (Odisha)	Adani LNG Pvt Ltd	5.00	-	-	-	-	-	-
Dhabol Expansion ²	GAIL(KLPL)	3.30	-	-	-	-	-	-
Jaigarh (Maharashtra)	H-Energy	4.00	-	-	-	-	-	-
Jafrabad (Gujarat)	Swan Energy	5.00	-	-	-	-	-	-
Chhara (Gujarat)	HSEPL	5.00	-	-	-	-	-	-

Note: 1. To increase to 5 MMTPA with breakwater. Only HP stream of capacity of 2.9 MMTPA is commissioned.

P:Provisional

2. Additional Capacity to be available after construction of breakwater

Source: CPSEs/Gas Project Division, MoPNG

Production of Crude Oil and Condensate (Calendar Year-wise)

State/Region	2016	2017	2018	2019	2020	2021 (P)
(1)	(2)	(3)	(4)	(5)	(6)	(7)
1. State-wise Crude Oil Production (TMT)*						
Andhra Pradesh	273	322	299	252	206	198
Arunachal Pradesh	58	52	43	52	55	50
Assam	4153	4341	4353	4113	3951	3949
Gujarat	4590	4597	4622	4676	4668	4646
Rajasthan	8288	7876	7832	7024	5859	5964
Tamil Nadu	269	341	375	415	413	383
Tripura	-	-	-	-	-	2
West Bengal	-	-	-	-	-	0.1
Offshore	18412	18369	17163	16111	15625	14699
Total	36042	35898	34687	32642	30777	29891
of which						
Crude Oil Production	34592	34302	32969	30959	29329	28622
Condensate Production	1450	1596	1718	1683	1448	1269
2. Entity-wise Crude Oil Production (TMT)*						
Onshore	17630	17529	17524	16531	15152	15192
of which						
ONGC-Nomination	5913	6016	6053	6064	5990	5848
OIL-Nomination	3206	3376	3344	3131	2979	2955
Pvt/JVs-PSC + RSC/CBM	8512	8137	8127	7336	6183	6389
Offshore	18412	18369	17163	16111	15625	14699
of which						
ONGC-Nomination	16199	16422	15256	14457	14459	13722
Pvt/JVs-PSC + RSC/CBM	2212	1947	1907	1653	1166	977
Total Crude Oil Production*	36042	35898	34687	32642	30777	29891
of which						
ONGC-Nomination	22112	22438	21309	20521	20449	19570
OIL-Nomination	3206	3376	3344	3131	2979	2955
Pvt/JVs-PSC + RSC/CBM	10724	10084	10035	8989	7350	7366

Note: Total may not tally due to rounding off

P: Provisional

*:Included Condensate

Source: Oil & Natural Gas Corporation Ltd., Oil India Ltd. and DGH

Petroleum Refining Capacity in India

(Figures in MMTPA)

Refinery Location & Year of Commissioning	2017	2018	2019	2020	2021	2022	Expansion Plan in next 5 Year	% Share of Total 2022
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Public Sector	144.97	148.07	149.87	150.37	149.72	151.72	184.12	60.4
1. IOCL, Digboi, Assam - 1901	0.65	0.65	0.65	0.65	0.65	0.65	0.65	0.3
2. IOCL, Guwahati, Assam - 1962	1.00	1.00	1.00	1.00	1.00	1.00	1.20	0.4
3. IOCL, Barauni, Bihar - 1964	6.00	6.00	6.00	6.00	6.00	6.00	9.00	2.4
4. IOCL, Koyali, Gujarat - 1965	13.70	13.70	13.70	13.70	13.70	13.70	18.00	5.5
5. IOCL, Haldia, West Bengal - 1975	7.50	7.50	7.50	8.00	8.00	8.00	8.00	3.2
6. Mathura, Uttar Pradesh - 1982	8.00	8.00	8.00	8.00	8.00	8.00	9.20	3.2
7. IOCL, Panipat, Haryana - 1998	15.00	15.00	15.00	15.00	15.00	15.00	25.00	6.0
8. IOCL, Bongaigaon, Assam - 1974	2.35	2.35	2.35	2.35	2.70	2.70	2.70	1.1
9. IOCL, Paradip, Odisha-2016	15.00	15.00	15.00	15.00	15.00	15.00	15.00	6.0
Indian Oil Corporation Ltd.	69.20	69.20	69.20	69.70	70.05	70.05	88.75	27.9
10. BPCL, Mumbai, Maharashtra - 1955	12.00	12.00	12.00	12.00	12.00	12.00	12.00	4.8
11. BPCL, Kochi, Kerala - 1963	12.40	15.50	15.50	15.50	15.50	15.50	15.50	6.2
12. BOREL, Bina, Madhya Pradesh - 2011	6.00	6.00	7.80	7.80	7.80	7.80	7.80	3.1
Bharat Petroleum Corporation Ltd.	30.40	33.50	35.30	35.30	35.30	35.30	35.30	14.1
13. HPCL, Mumbai, Maharashtra - 1954	7.50	7.50	7.50	7.50	7.50	9.50	7.50	3.8
14. HPCL, Visakh, Andhra Pradesh - 1957	8.30	8.30	8.30	8.30	8.30	8.30	15.00	3.3
Hindustan Petroleum Corporation Ltd.	15.80	15.80	15.80	15.80	15.80	17.80	22.50	7.1
15. CPCL, Manali, Tamil Nadu - 1965	10.50	10.50	10.50	10.50	10.50	10.50	10.50	4.2
16. CPCL- CBR, Narimanam Tamil Nadu - 1993	1.00	1.00	1.00	1.00	0.00	0.00	9.00	0.0
Chennai Petroleum Corporation Ltd.	11.50	11.50	11.50	11.50	10.50	10.50	19.50	4.2
17. NRL, Numaligarh, Assam-1999	3.00	3.00	3.00	3.00	3.00	3.00	3.00	1.2
18. ONGC, Tatipaka, Andhra Pradesh - 2001	0.07	0.07	0.07	0.07	0.07	0.07	0.07	0.0
19. MRPL, Mangalore, Karnataka- 1996	15.00	15.00	15.00	15.00	15.00	15.00	15.00	6.0
Private & JVs Sector	89.00	99.50	99.50	99.50	99.50	99.50	99.50	39.6
20. RIL, Jamnagar, Gujarat - 1999	33.00	33.00	33.00	33.00	33.00	33.00	33.00	13.1
21. RIL, Jamnagar (SEZ), Gujarat - 2008	27.00	35.20	35.20	35.20	35.20	35.20	35.20	14.0
Reliance Industries Ltd. (RIL)	60.00	68.20	68.20	68.20	68.20	68.20	68.20	27.1
22. NEL, Vadinar, Gujarat - 2006	20.00	20.00	20.00	20.00	20.00	20.00	20.00	8.0
23. HMEL, Bhatinda, Punjab - 2012	9.00	11.30	11.30	11.30	11.30	11.30	11.30	4.5
Total Refinery Capacity	233.97	247.57	249.37	249.87	249.22	251.22	283.62	100.0

Consumption of Natural Gas

(Figures in MMSCM)

Sector	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22 (P)
(1)	(2)	(3)	(4)	(5)	(6)	(7)
(a) Energy Purpose						
Power	11616	12028	12005	11080	10836	10157
Industrial & Manufacturing	794	999	1086	701	555	829
City or Local Natural Gas Distribution Network	7350	8585	9206	10883	9230	12175
Agriculture (Tea Plantation)	183	189	192	200	177	156
Internal Consumption for Pipeline System	471	496	541	525	439	486
Refinery	5374	6533	7047	7786	7911	5312
LPG Shrinkage	759	798	874	858	900	1070
Miscellaneous	3929	3226	3393	4209	4569	9229
Total (a)	30478	32854	34343	36241	34617	39414
(b) Non-Energy Purpose						
Fertilizer Industry	15429	14676	14987	16115	17781	18079
Petrochemical	4170	4024	3386	3569	3072	2864
Sponge Iron	885	1278	1124	567	647	1134
Total (b)	20484	19978	19497	20251	21500	22077
Total Sectoral Sales (a+b)	50961	52832	53840	56492	56117	61491
Total Consumption **	55697	59170	60796	64144	60815	63907
Total Consumption in MMSCMD	153	162	167	175	167	175

Oil & Gas Import Dependency

Elements	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22 (P)
(1)	(2)	(3)	(4)	(5)	(6)	(7)
Import dependence on crude oil (in MMT)						
Total Consumption	194.6	206.2	213.2	214.1	194.3	204.2
Indigenous crude oil processing	33.5	32.8	31.7	29.3	28.0	27.0
Products from Indigenous crude	31.3	30.6	29.6	27.3	26.1	25.2
Products from Fractionators	4.3	4.6	4.9	4.8	4.2	4.1
Total Indigenous production	35.6	35.2	34.5	32.1	30.3	29.3
Self Sufficiency (%)	18.3	17.1	16.2	15.0	15.6	14.3
Import Dependence (%)	81.7	82.9	83.8	85.0	84.4	85.7
Import dependence on natural gas (in BCM)						
Total Consumption	55.7	59.2	60.8	64.1	60.8	63.9
Indigenous gas production	30.8	31.7	32.1	30.3	27.8	33.1
Import of LNG	24.8	27.4	28.7	33.9	33.0	30.8
Self Sufficiency (%)	55.4	53.6	52.7	47.2	45.7	51.8
Import Dependence (%)	44.6	46.4	47.3	52.8	54.3	48.2
Import dependence on oil and oil equivalent gas (O+OEG) (in MMTOE)						
Total Consumption	264.3	280.2	289.3	293.6	269.1	282.8
Indigenous O+OEG production	68.9	69.4	69.1	64.6	60.3	64.5
Self Sufficiency (%)	26.1	24.8	23.9	22.0	22.4	22.8
Import Dependence (%)	73.9	75.2	76.1	78.0	77.6	77.2

Consumption of Various Energy Sources in India

Year	Coal	Lignite	Total Coal & Lignite	Natural Gas	Crude Oil	Petroleum Products	Electricity
	(TMT)			(MMSCM)	(TMT)		(GWh)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
2016-17	837.22	43.16	880.38	55.70	245.36	194.60	1061.18
2017-18	898.52	46.32	944.84	59.17	251.93	206.17	1123.43
2018-19	968.25	45.81	1014.06	60.80	257.20	213.22	1209.97
2019-20	950.11	42.32	992.42	64.14	254.39	214.13	1248.09
2020-21	906.13	38.51	944.64	60.82	221.77	194.30	1230.21
2021-22 (P)	1027.93	49.14	1077.07	65.04	241.70	204.23	1291.73

Availability of Various Energy Sources in India

Year	Coal	Lignite	Total Coal & Lignite	Natural Gas	Crude Oil	Petroleum Products	Electricity
	(TMT)			(MMSCM)	(TMT)		(GWh)
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
2016-17	858.58	47.32	905.89	55.70	249.94	214.32	1163.29
2017-18	896.09	46.98	943.06	59.17	256.12	223.03	1232.51
2018-19	958.25	42.72	1000.97	60.80	260.70	234.61	1307.68
2019-20	1001.84	42.23	1044.07	64.14	259.12	241.04	1323.05
2020-21	956.02	36.21	992.23	60.82	226.95	219.99	1314.02
2021-22 (P)	945.78	45.95	991.73	65.04	241.67	233.65	1371.37

Thank you